

Homework 3 Solutions

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October 2025

Problem 1

Let x, y, z denote the numbers of chickens, cows, and pigs respectively. The system is

$$\begin{aligned}
 & \left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 2 & 4 & 4 & 140 \\ 1 & -2 & 0 & 0 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 0 & 2 & 2 & 40 \\ 1 & -2 & 0 & 0 \end{array} \right] \xrightarrow{R_3 \leftarrow R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 0 & 2 & 2 & 40 \\ 0 & -3 & -1 & -50 \end{array} \right] \\
 & \xrightarrow{R_2 \leftarrow \frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 0 & 1 & 1 & 20 \\ 0 & -3 & -1 & -50 \end{array} \right] \xrightarrow{R_3 \leftarrow R_3 + 3R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 0 & 1 & 1 & 20 \\ 0 & 0 & 2 & 10 \end{array} \right] \xrightarrow{R_3 \leftarrow \frac{1}{2}R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 0 & 1 & 1 & 20 \\ 0 & 0 & 1 & 5 \end{array} \right] \\
 & \xrightarrow{R_2 \leftarrow R_2 - R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 0 & 1 & 0 & 15 \\ 0 & 0 & 1 & 5 \end{array} \right] \xrightarrow{R_1 \leftarrow R_1 - R_3} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 45 \\ 0 & 1 & 0 & 15 \\ 0 & 0 & 1 & 5 \end{array} \right] \xrightarrow{R_1 \leftarrow R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 30 \\ 0 & 1 & 0 & 15 \\ 0 & 0 & 1 & 5 \end{array} \right].
 \end{aligned}$$

Thus the reduced row-echelon form has the identity matrix on the left, giving

$$x = 30, y = 15, z = 5.$$

Problem 2

System:

$$\begin{aligned}
 x + 2y - z &= 1, \\
 2x + 4y - 2z &= 2, \\
 3x + 6y - 3z &= 5.
 \end{aligned}$$

Augmented matrix and elimination:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 4 & -2 & 2 \\ 3 & 6 & -3 & 5 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1, R_3 \leftarrow R_3 - 3R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{array} \right].$$

The last row reads $0x + 0y + 0z = 2$, which is impossible. Hence, the system is inconsistent and has no solutions.

Problem 3

A is $3 \times k$ and B is 2×4 . For AB to be defined, the inner dimensions must match (cols in A = rows in B), so $k = 2$. The product then has size 3×4 .

$$\boxed{k = 2, \quad AB \in \mathbb{R}^{3 \times 4}}.$$

Problem 4

Let $\mathbf{u} = (1, 2)$ and $\mathbf{v} = (3, 4)$.

$$\mathbf{u} \cdot \mathbf{v} = 1 \cdot 3 + 2 \cdot 4 = 11, \quad \|\mathbf{u}\| = \sqrt{1^2 + 2^2} = \sqrt{5}, \quad \|\mathbf{v}\| = \sqrt{3^2 + 4^2} = 5.$$

Thus

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{11}{5\sqrt{5}}, \quad \theta = \arccos\left(\frac{11}{5\sqrt{5}}\right) \approx 10.3^\circ \approx 0.18 \text{ Radians}.$$

Problem 5

(a)

$$\begin{bmatrix} 5 & -4 & -2 \\ 5 & -5 & 4 \\ 2 & 5 & -4 \\ -5 & 4 & 3 \\ 3 & -4 & -3 \end{bmatrix} \begin{bmatrix} 5 \\ -2 \\ -3 \end{bmatrix}$$

Is defined and will result in a 5×1 matrix:

$$\begin{bmatrix} 5 & -4 & -2 \\ 5 & -5 & 4 \\ 2 & 5 & -4 \\ -5 & 4 & 3 \\ 3 & -4 & -3 \end{bmatrix} \begin{bmatrix} 5 \\ -2 \\ -3 \end{bmatrix} = \begin{bmatrix} (5)(5) + (-4)(-2) + (-2)(-3) \\ (5)(5) + (-5)(-2) + 4(-3) \\ (2)(5) + (5)(-2) + (-4)(-3) \\ (-5)(5) + (4)(-2) + 3(-3) \\ (3)(5) + (-4)(-2) + (-3)(-3) \end{bmatrix} = \begin{bmatrix} 39 \\ 23 \\ 12 \\ -42 \\ 32 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 0 & -1 & -1 & 3 \\ 5 & -5 & -2 & 2 \\ 1 & 0 & 4 & 5 \end{bmatrix} \begin{bmatrix} 0 & -3 \\ -2 & -1 \\ 3 & -3 \end{bmatrix}$$

Not defined: the left matrix is 3×4 and the right matrix is 3×2 ; the inner dimensions (4 and 3) do not match, so the product is not defined.